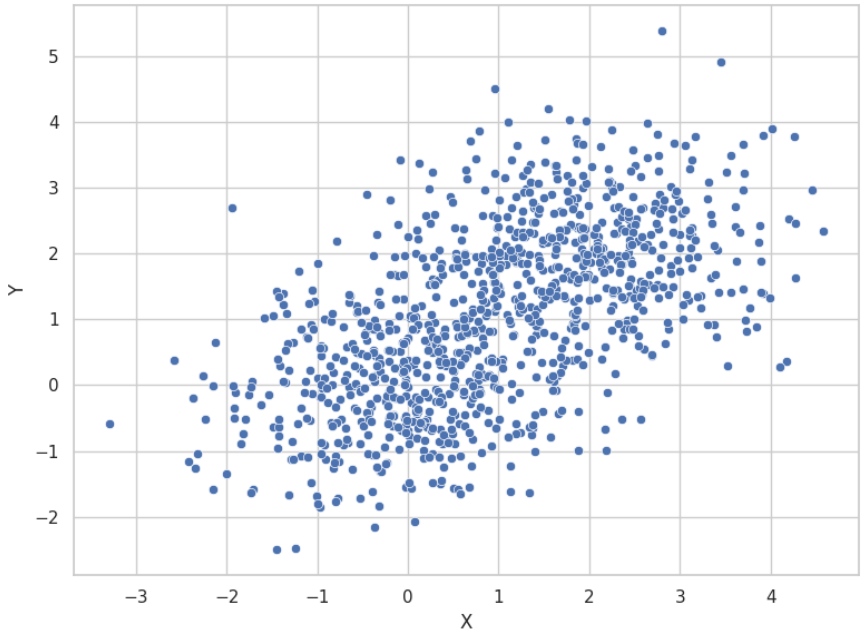
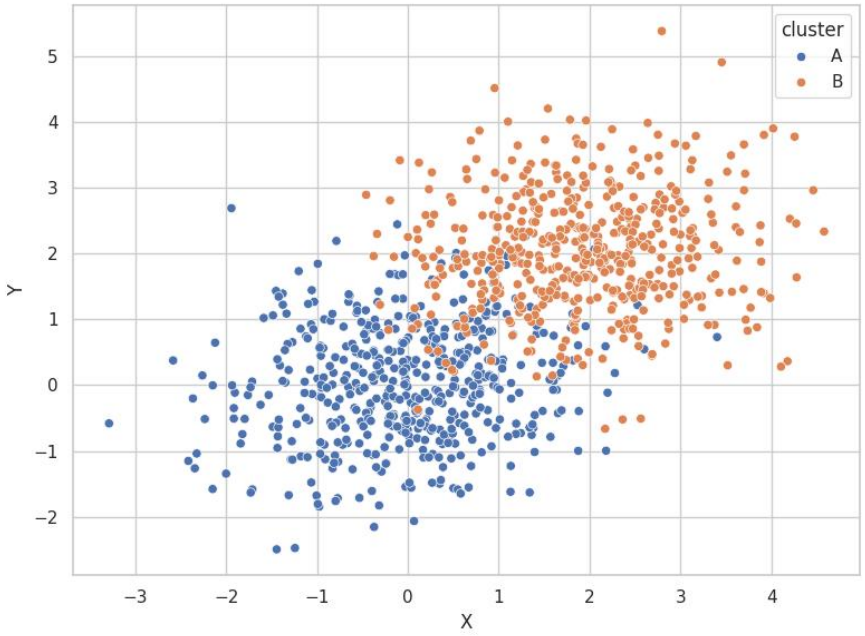
03

Regression

Regression vs classification





REGRESSION

Output = continuous

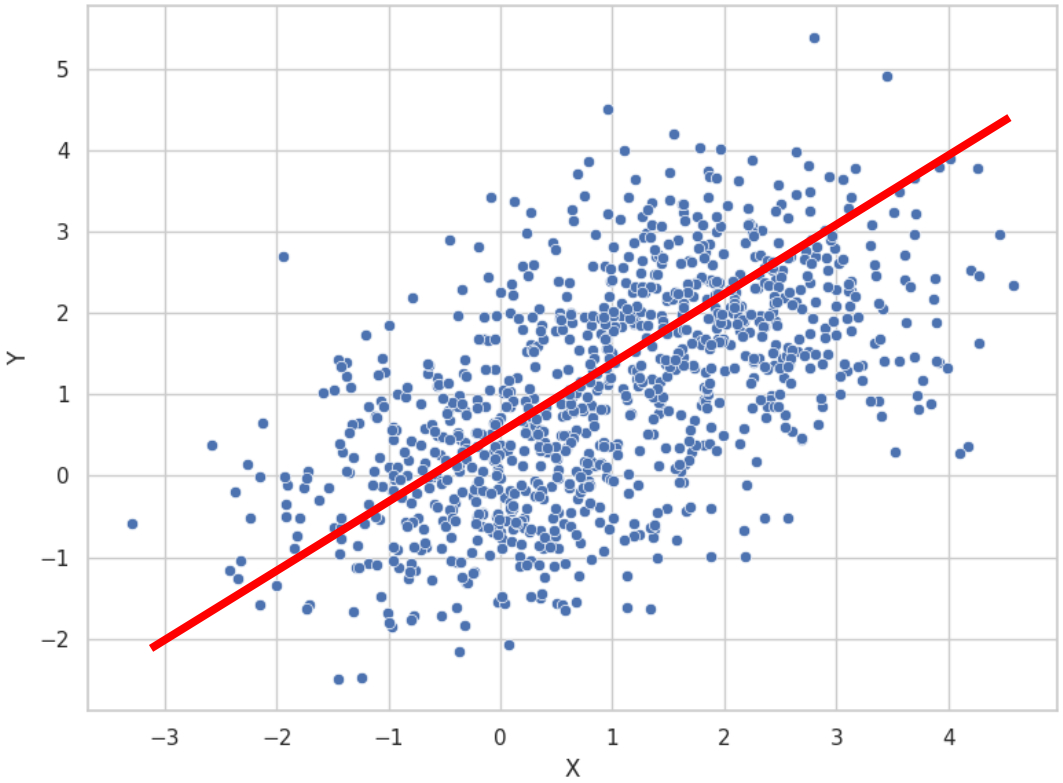
Based on X, predict the value of Y

CLASSIFICATION

Output = discrete

Based on X and Y, predict the class

Linear regression



We need to make

assumptions

about the

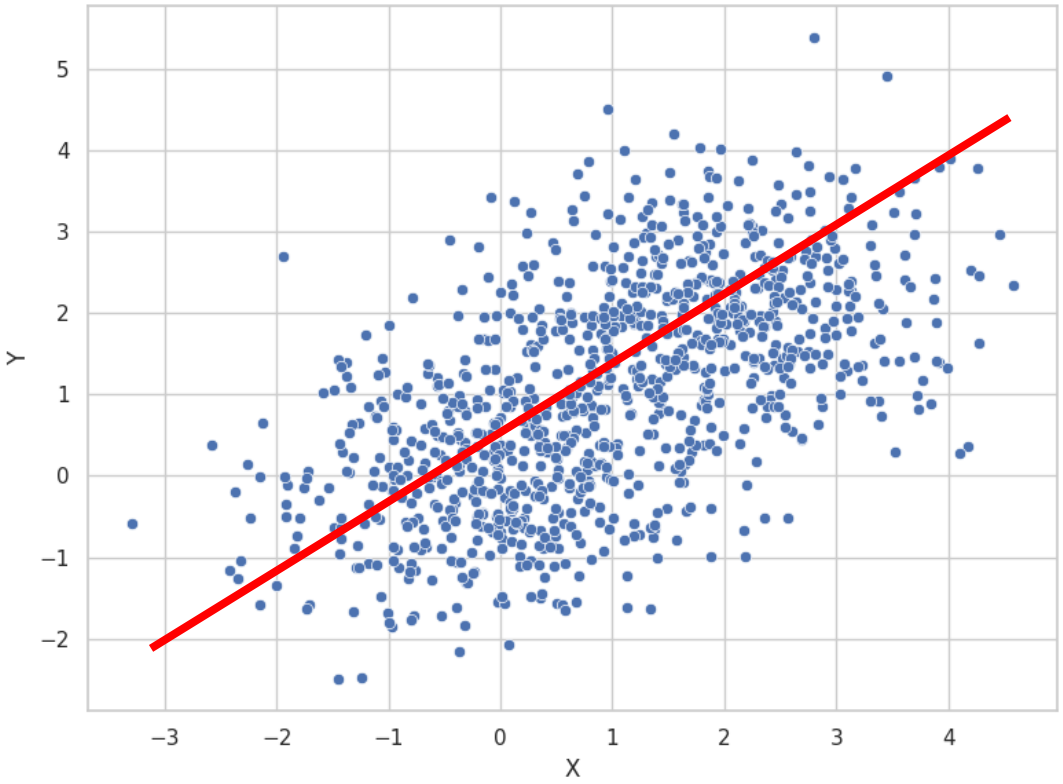
model

that generated the data

Linear relationship

Linear model

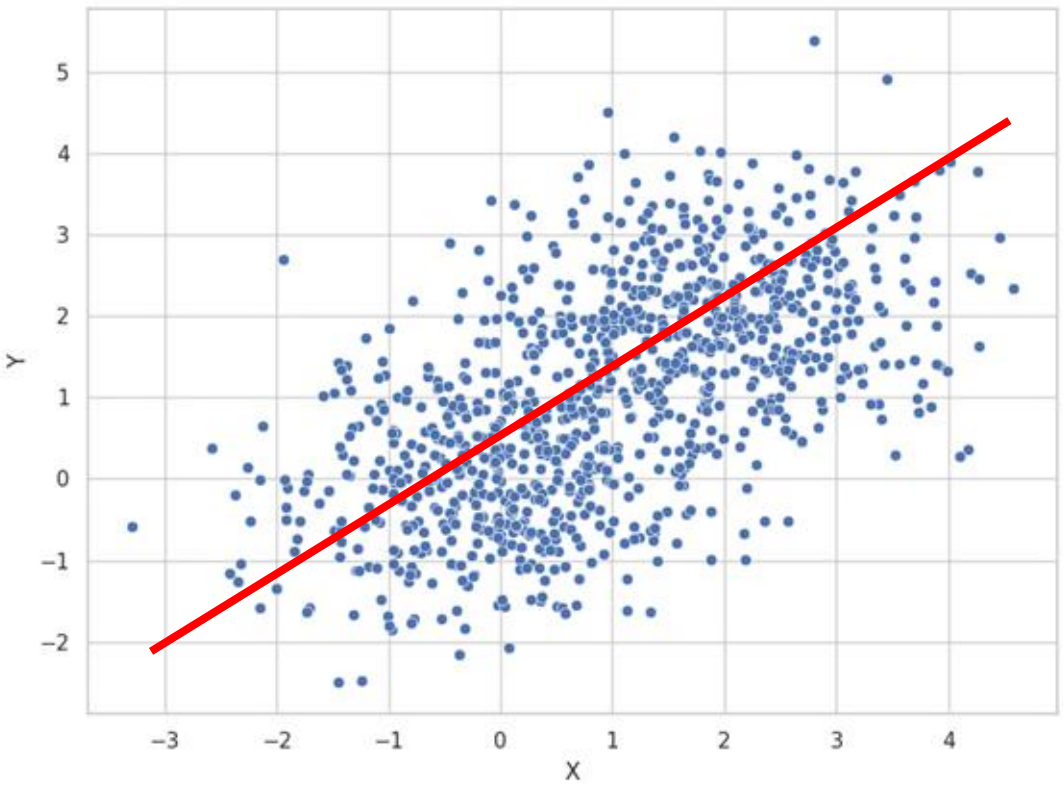
Linear regression



- Linear equation: Y = w*X + b
- Model parameters
  - w = weight (slope)
  - b = intercept (bias)

How do we optimize/evaluate our model?

Linear regression



STATISTICS

"What is the relationship between X and Y? Can we explain it?"

→ goodness-of-fit (R^2)

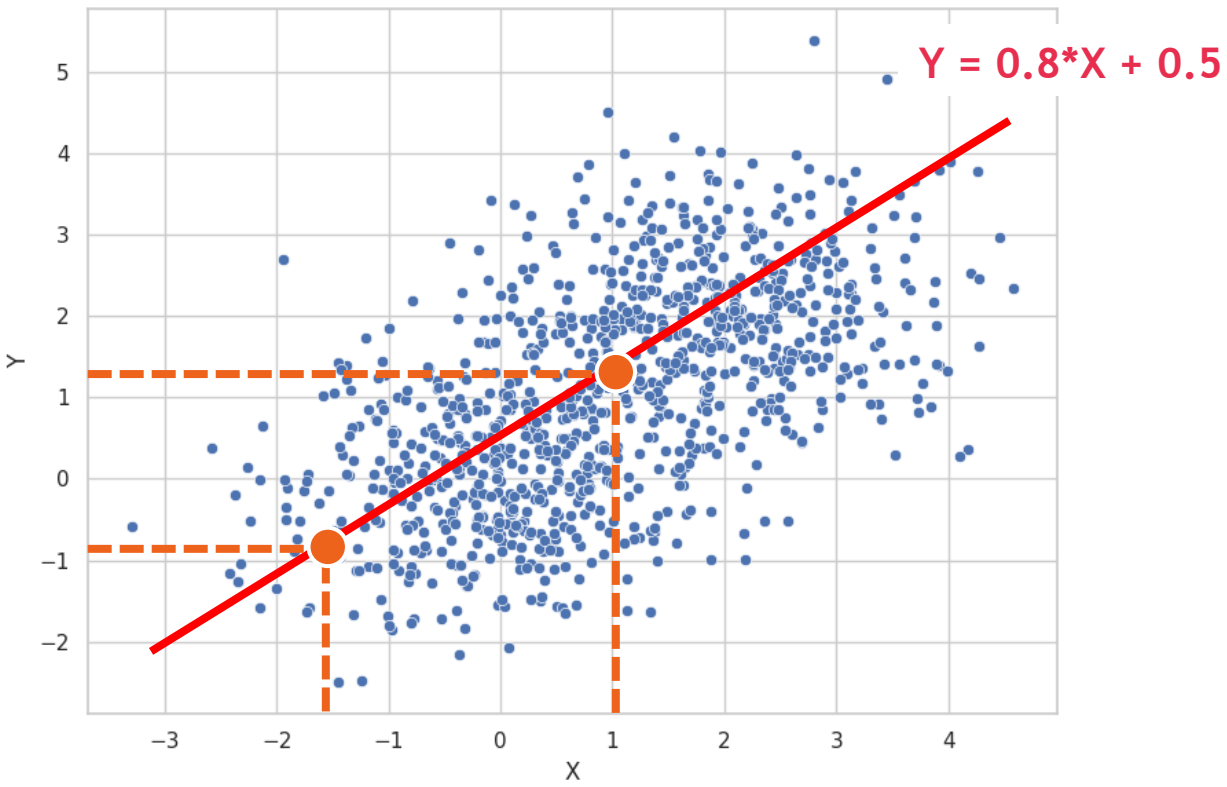
MACHINE LEARNING

"Given X, can we predict Y accurately?"

→ Predictive performance on test data

VIB

Linear regression - prediction

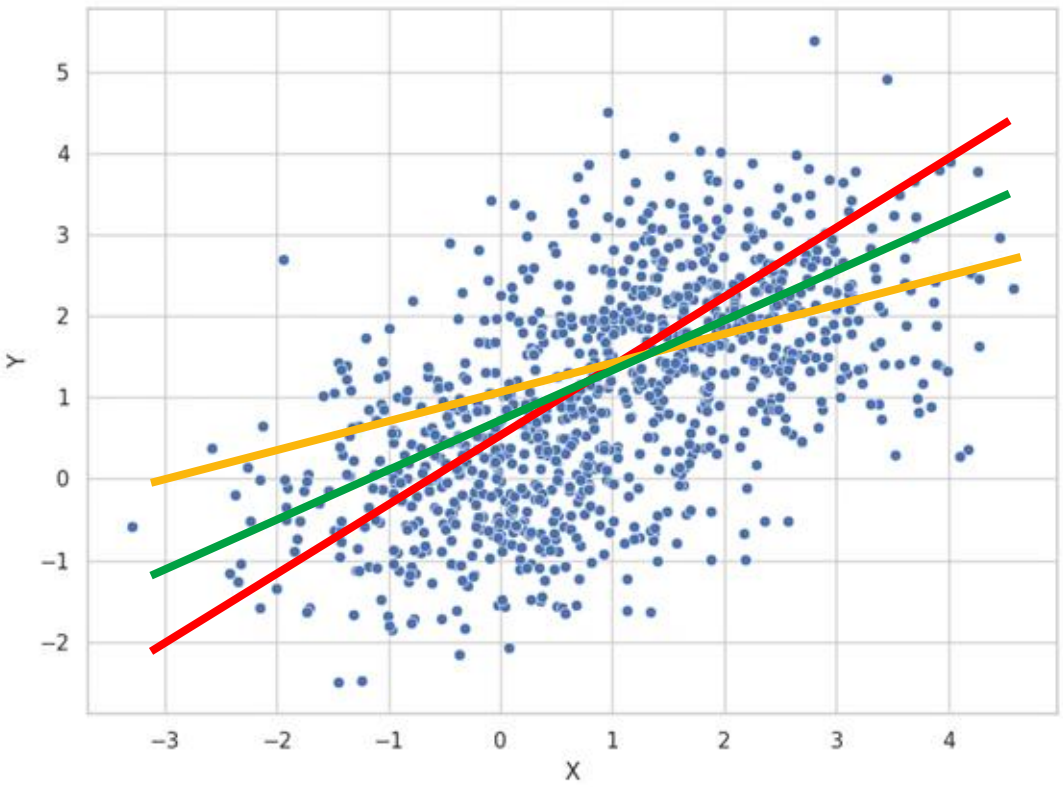


- Linear equation: Y = w*X + b
- Model parameters
  - w = weight (slope)
  - b = intercept (bias)
- Predictions (**generalization!**)

Linear regression - prediction

- Linear equation: Y = w*X + b
- Model parameters
  - w = weight (slope)
  - b = intercept (bias)
- Predictions (**generalization!**)

Linear regression - fitting

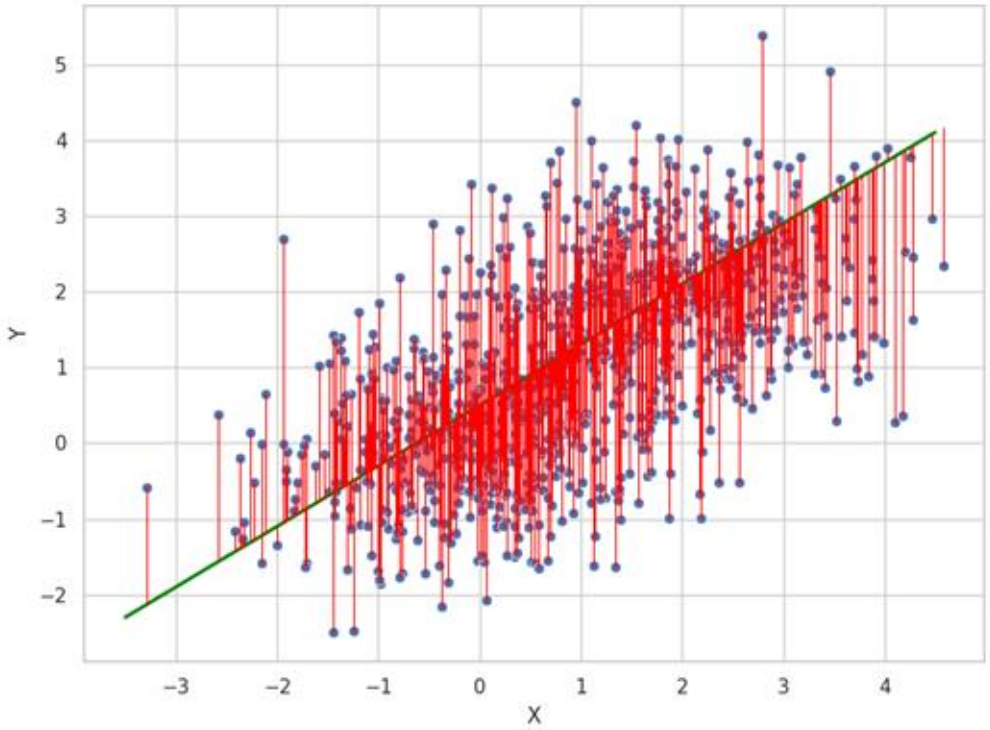


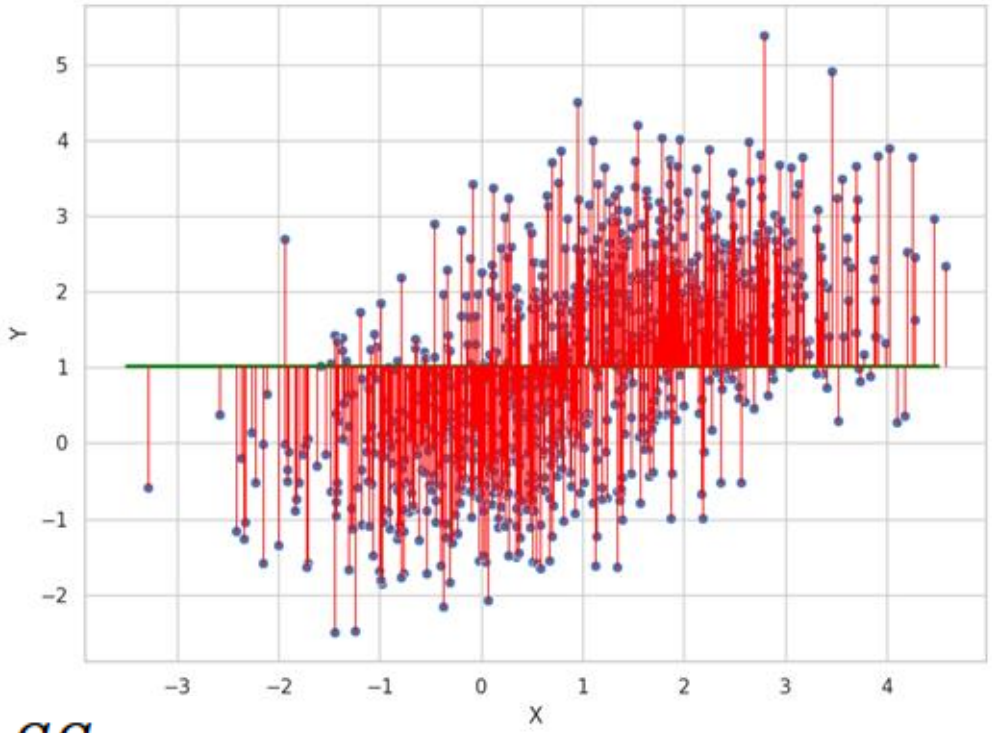
What values for *w* and *b* are optimal and give the best fit to the model?

Linear regression - fitting

$$SS_{error} = \sum_{i=1}^{n} (y^{(i)} - f(x^{(i)}))^2$$

$$SS_{total} = \sum_{i=1}^{n} (y^{(i)} - \bar{y})^2$$

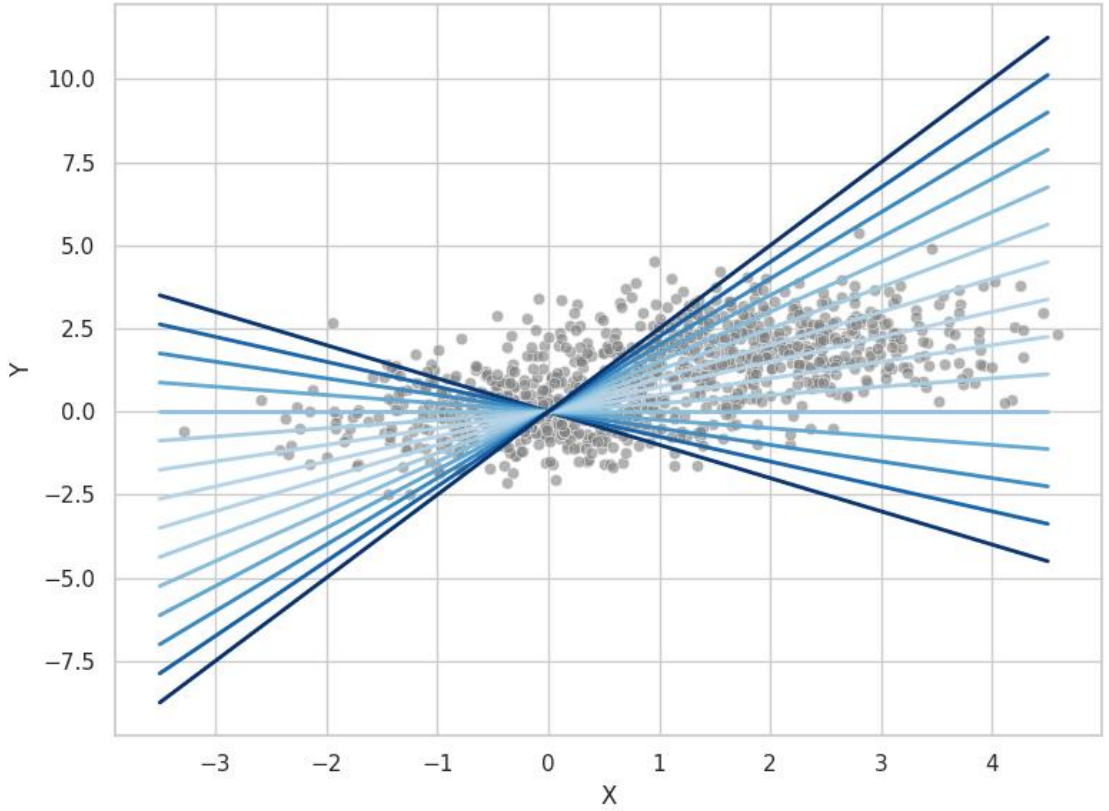


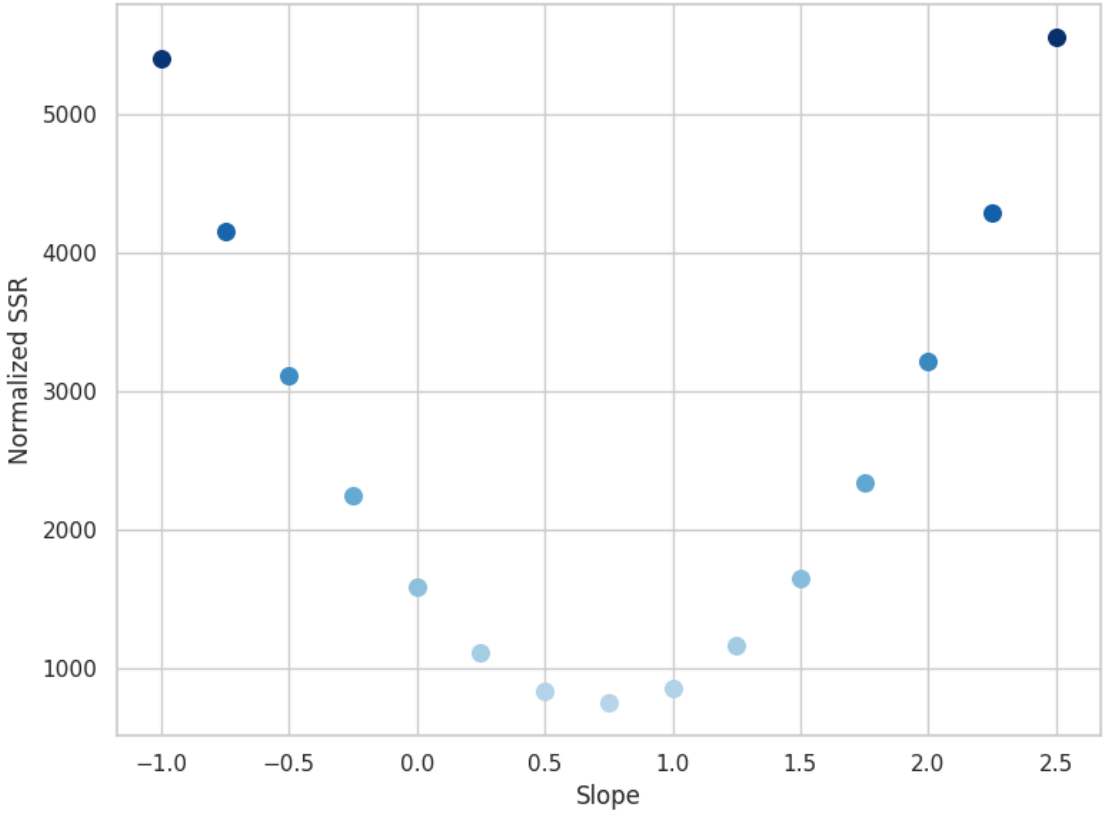


$$R^2 = 1 - \frac{SS_{error}}{SS_{total}}$$

Linear regression - fitting

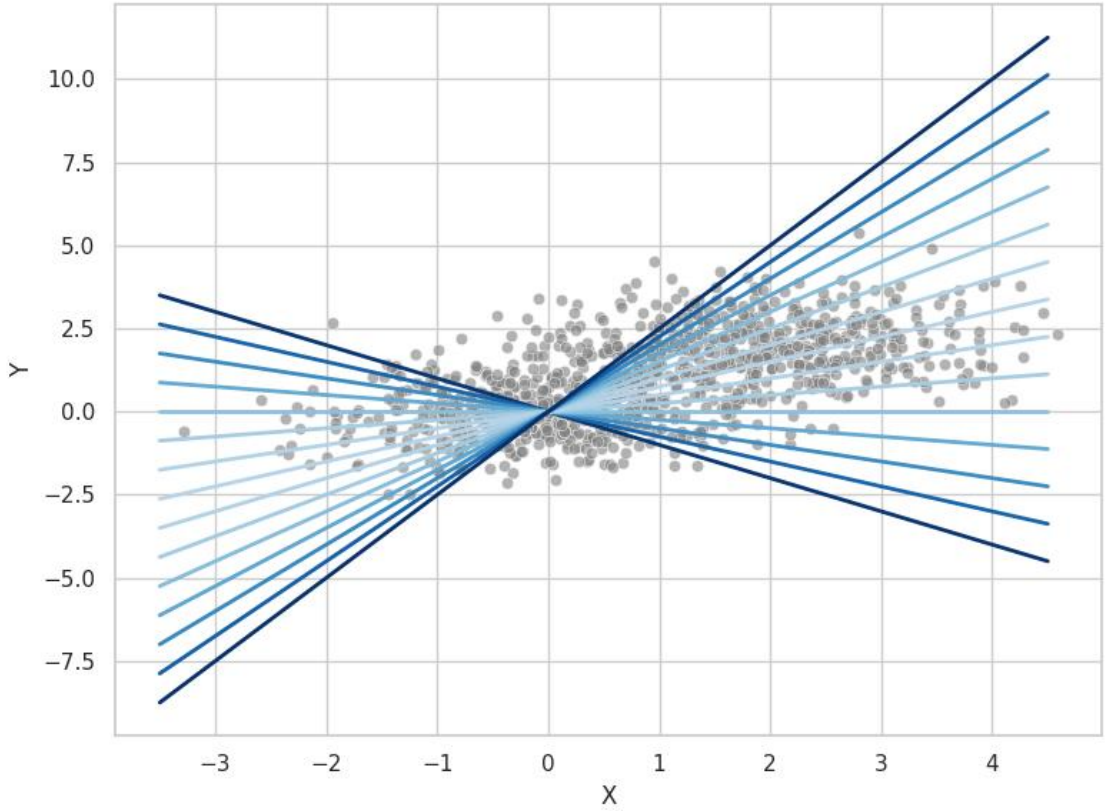
→ Finding the “best” slope

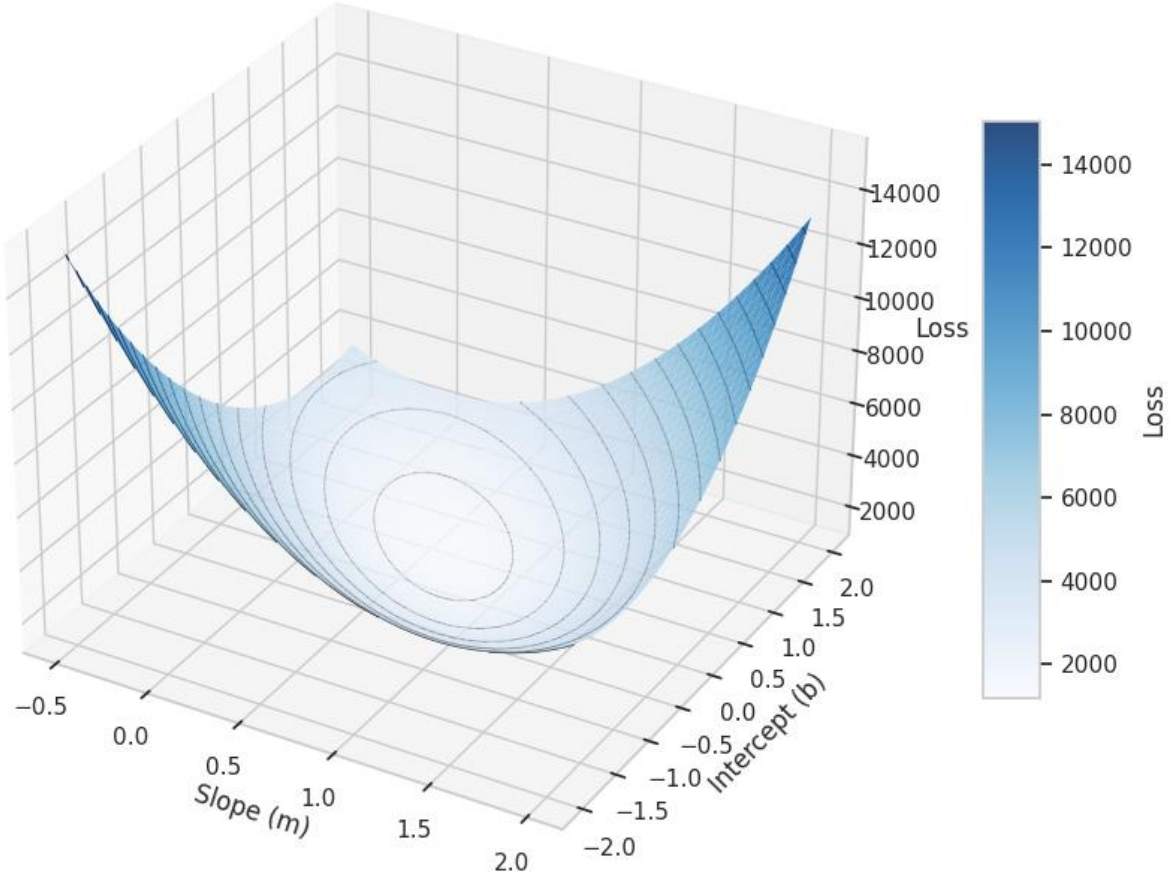




Linear regression - fitting

→ Finding the “best” combination of slope and intercept





Linear regression - fitting

Fit a linear model

$$f(x, \theta) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m + b$$

Linear regression - fitting

Fit a linear model

$$f(x,\theta) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m + b$$

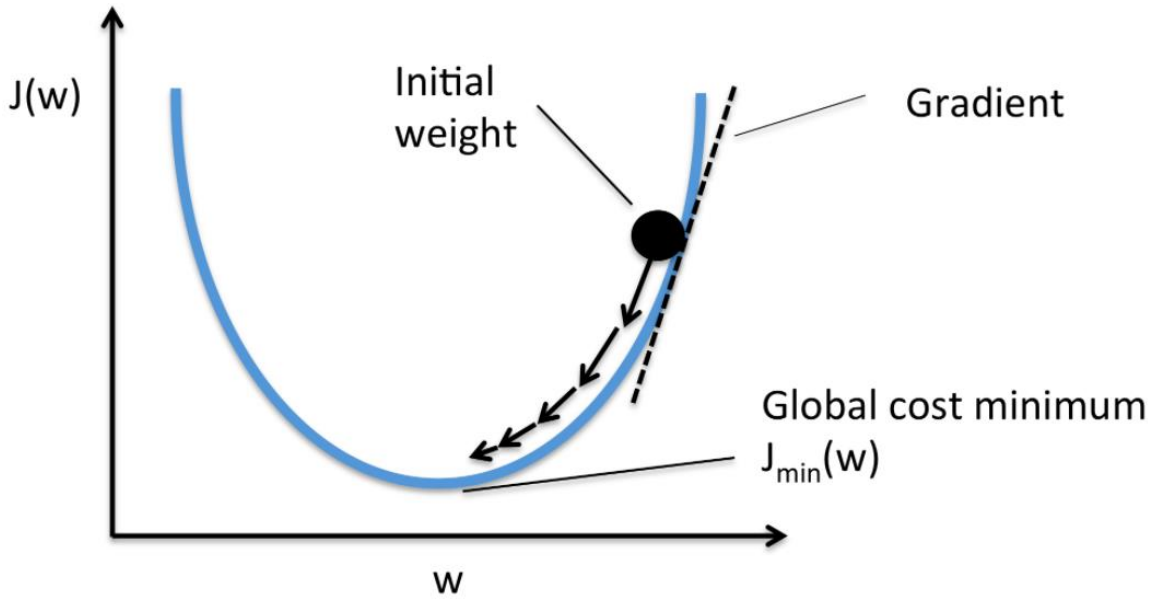
to the dataset such that the cost function

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}(f(x,\theta) - y)^2$$

is minimal

VIB

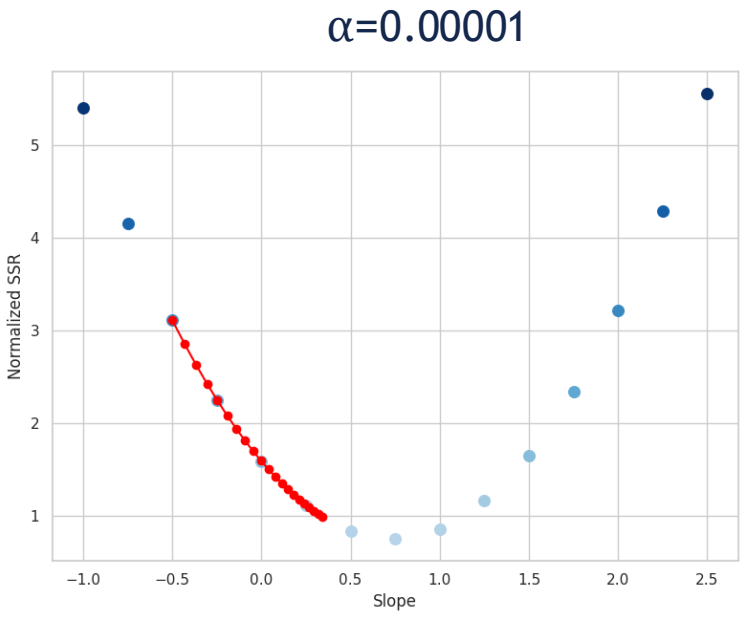
Gradient descent

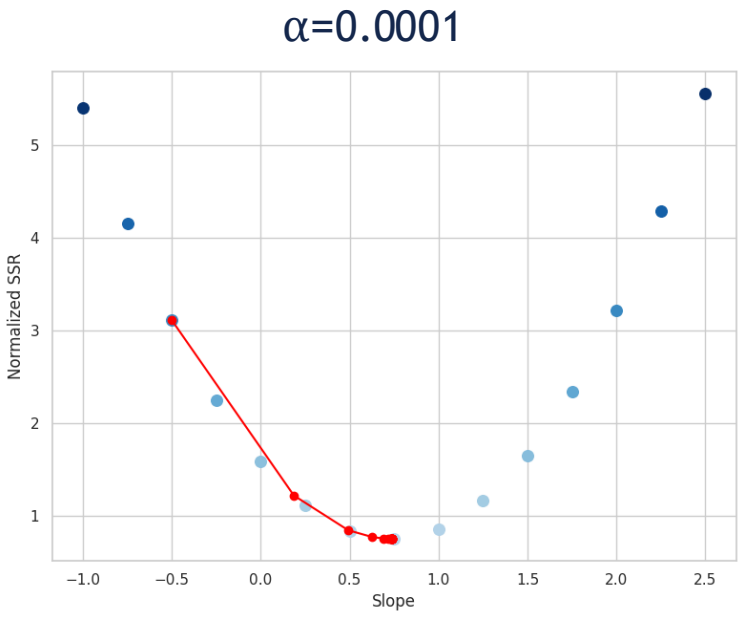


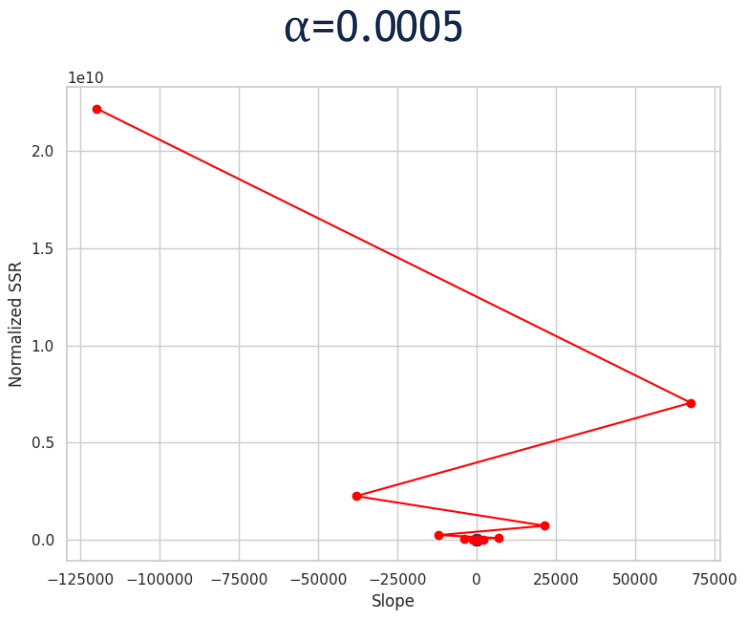
- optimization algorithm
  - **minimize a loss function**
  - **iteratively updating** model parameters
- direction of the weight change
  → the steepest descent = **negative gradient**

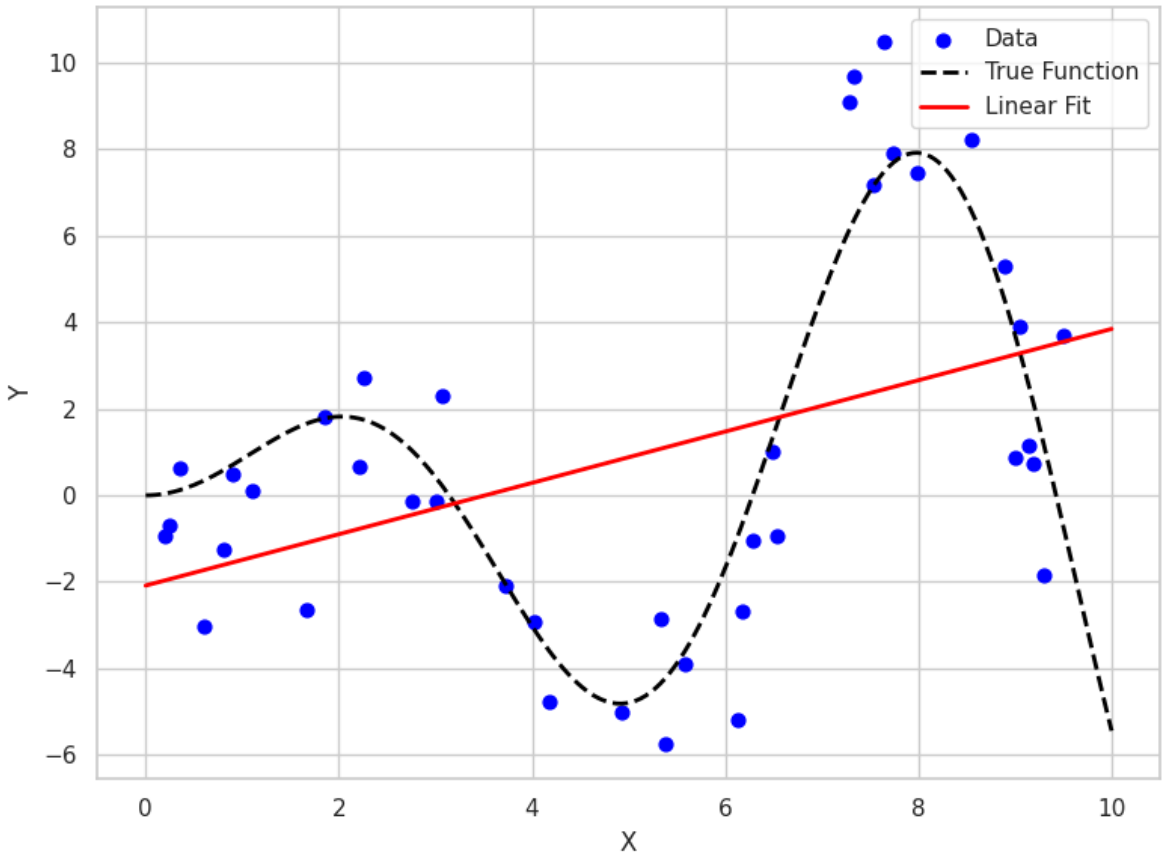
The **learning rate (α)** controls the step size of updates

Gradient descent

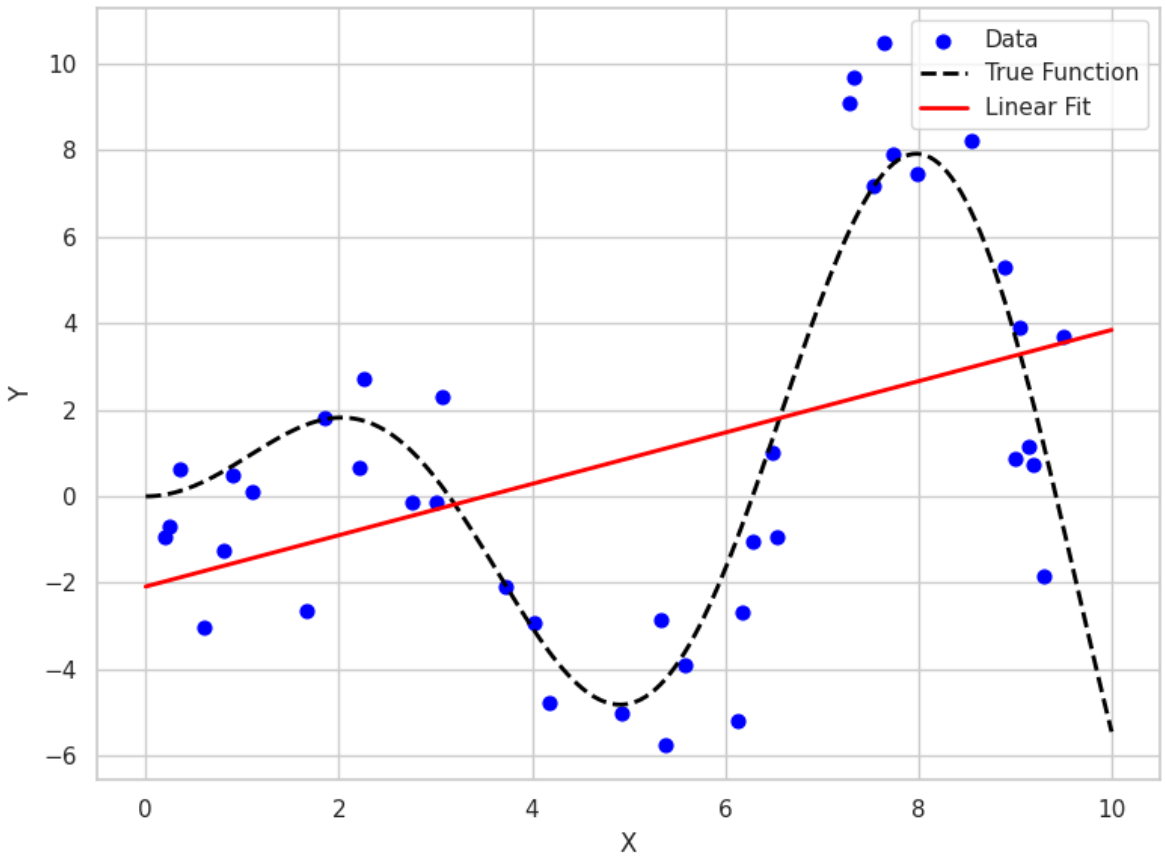







Data
True Function
Linear Fit
10
8
6
4
2
0
−2
−4
−6
Y
0
2
4
6
8
10
X

Non-linear regression



We need to make

assumptions

about the

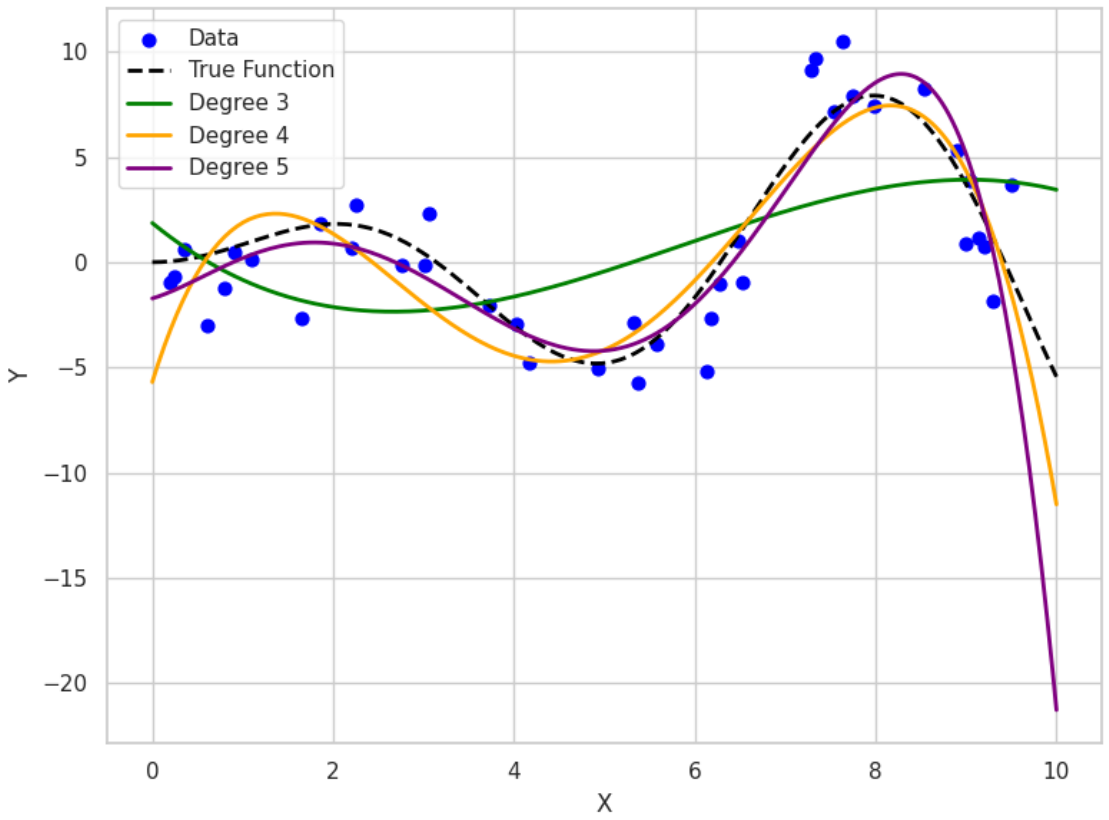
model

that generated the data

Non-linear relationship

Non-linear model

Non-linear regression



Polynomial transformation

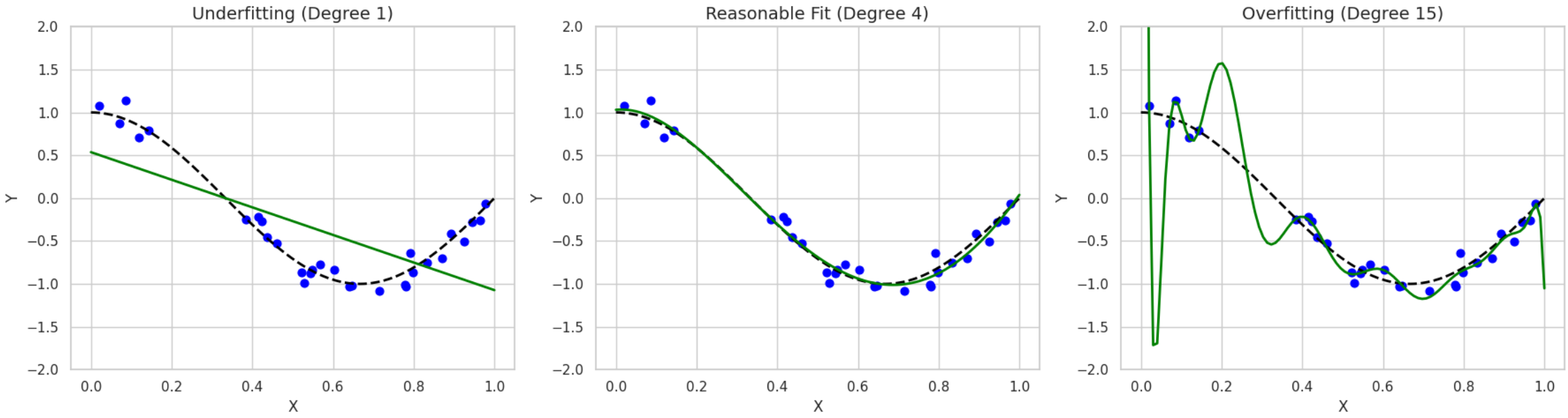
- Include higher-order features
- Allows to fit a linear model to non-linear relationships in the data

$$f(x, \theta) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \cdots$$

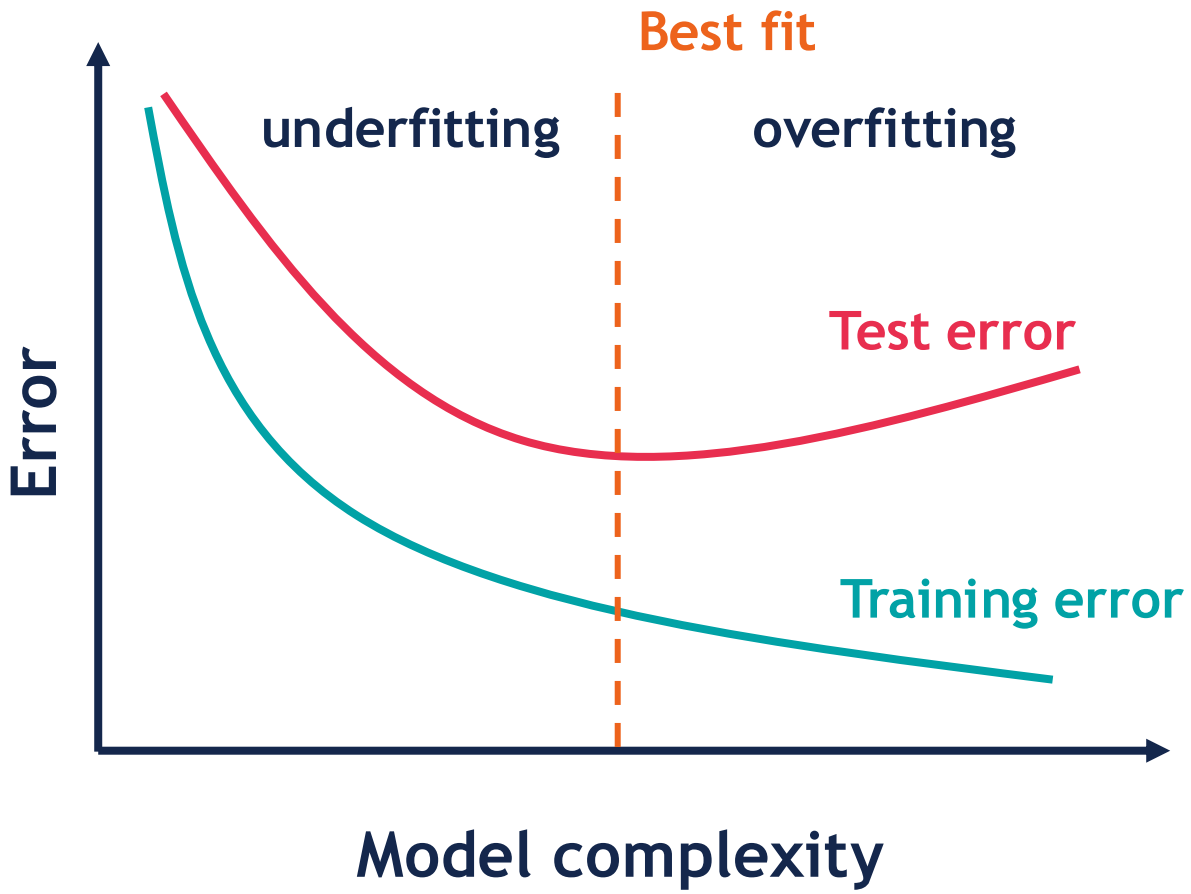
Increasing the polynomial degree
→ allows to fit more complex relationships

But …
→ Requires more data (more model parameters)
→ Can lead to poor generalization (overfitting)

Underfitting & overfitting



Train-test splitting



How can we limit the model complexity?

Model regularization

Every dataset contains noise → Label noise / Feature noise

Don’t try to fit the data perfectly!

Limit the model complexity

- Skip irrelevant/correlated features (feature selection)
- Choose a simple model (linear vs non-linear)
- Put a constraint on the model complexity (regularization)
- Check the generalizability with a validation/test set

Model regularization

Prevents overfitting by adding a **penalty term** to the loss function, discouraging overly complex models

L1 regularization (Lasso)

- Penalty term = sum of the absolute values of the model weights
- Promotes sparsity (weights are set to zero)

L2 regularization (Ridge)

- Penalty term = sum of the squared values of the model weights
- Prevents large weights and ensures smooth generalization

Elastic Net

- Combination of L1 and L2 penalties
- Balances feature selection and weight shrinkage
- Useful for high-dimensional datasets with correlated features

Regularized linear regression

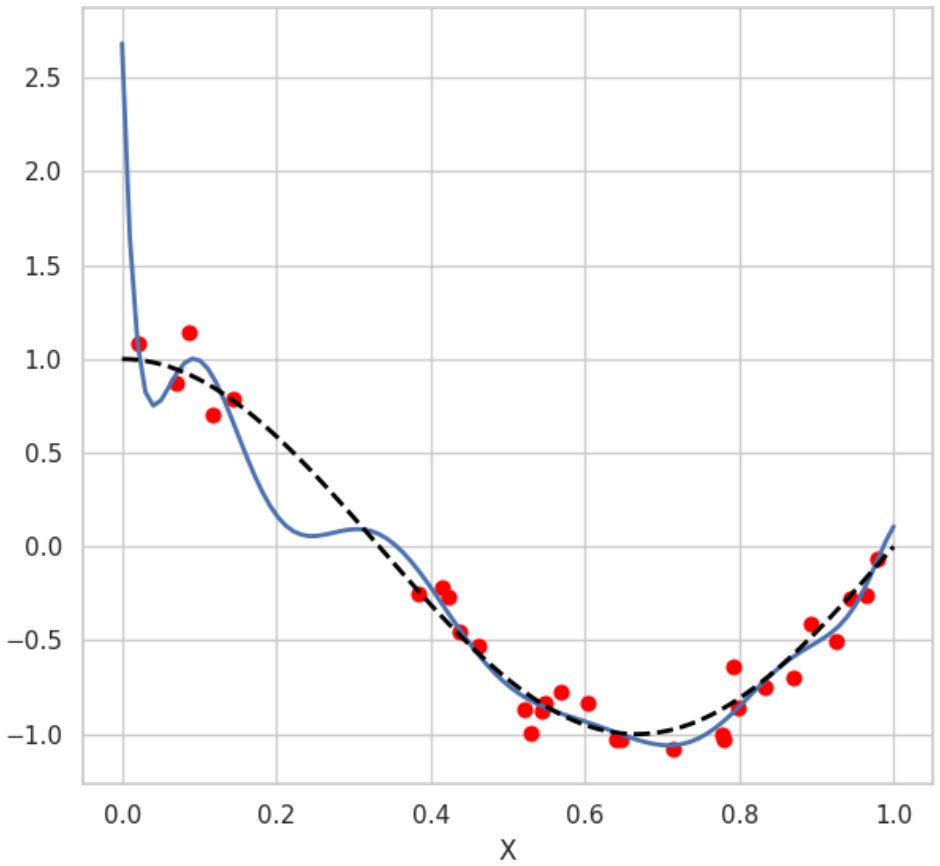
linear model

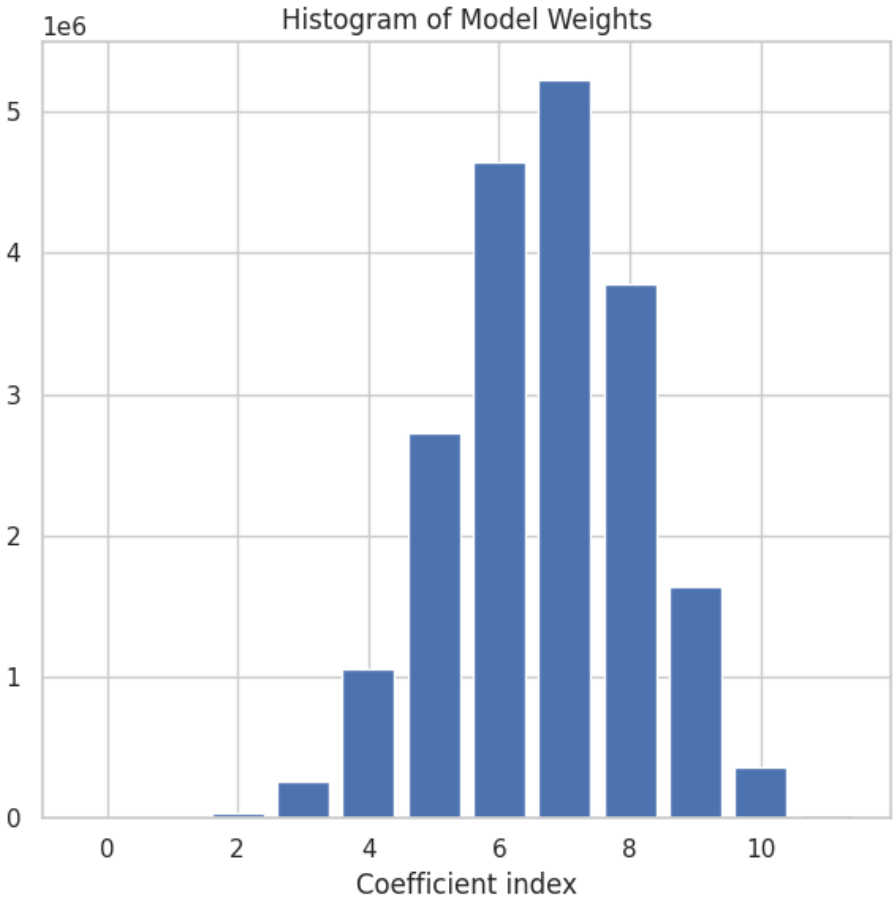
$$f(x,\theta) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m$$

$$J(\theta) = \underbrace{\frac{1}{2n}\sum_{i=1}^{n}\left(f\left(x^{(i)},\theta\right) - y^{(i)}\right)^2}_{\text{Cost function}} + \underbrace{\lambda_1 \sum_{j=1}^{m}\left|\theta_j\right|}_{\text{L1}} + \underbrace{\lambda_2 \sum_{j=1}^{m}\theta_j^2}_{\text{L2}}$$

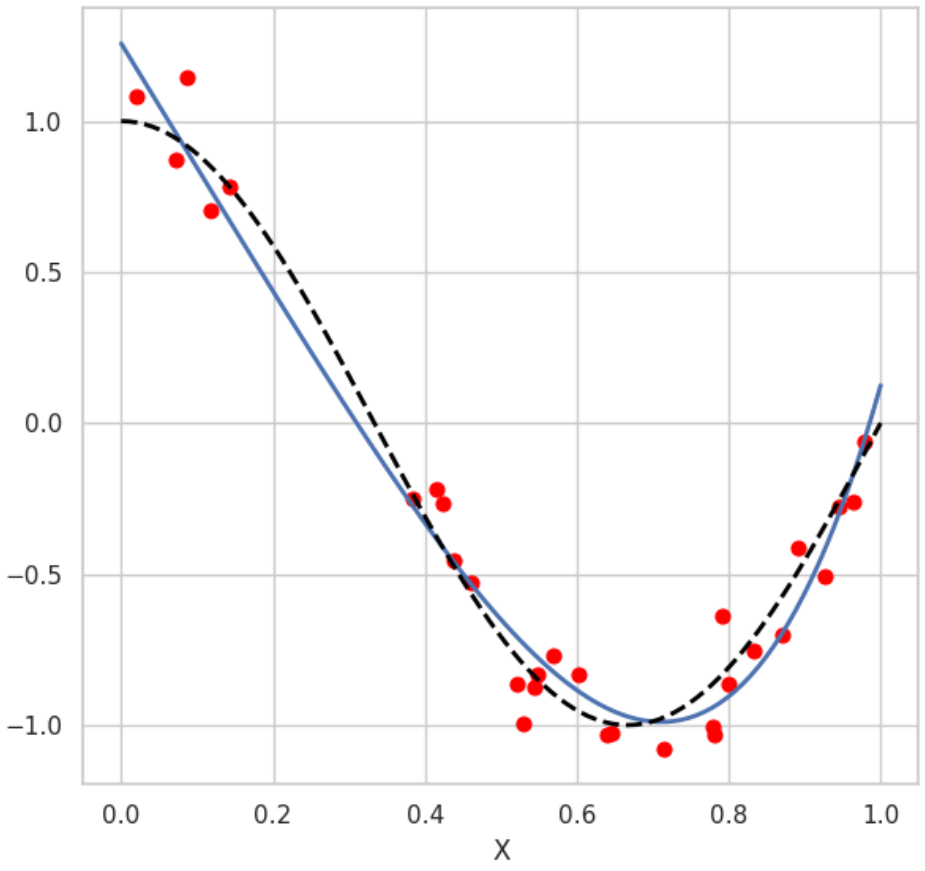
VIB

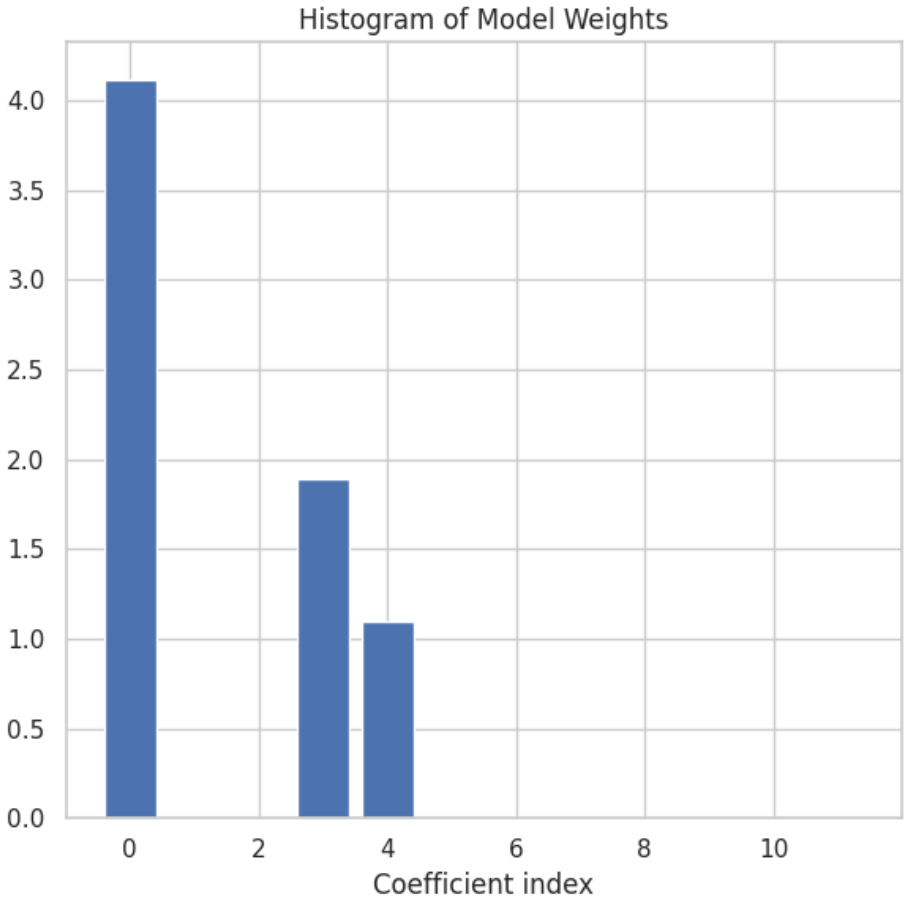
Non-regularized linear regression



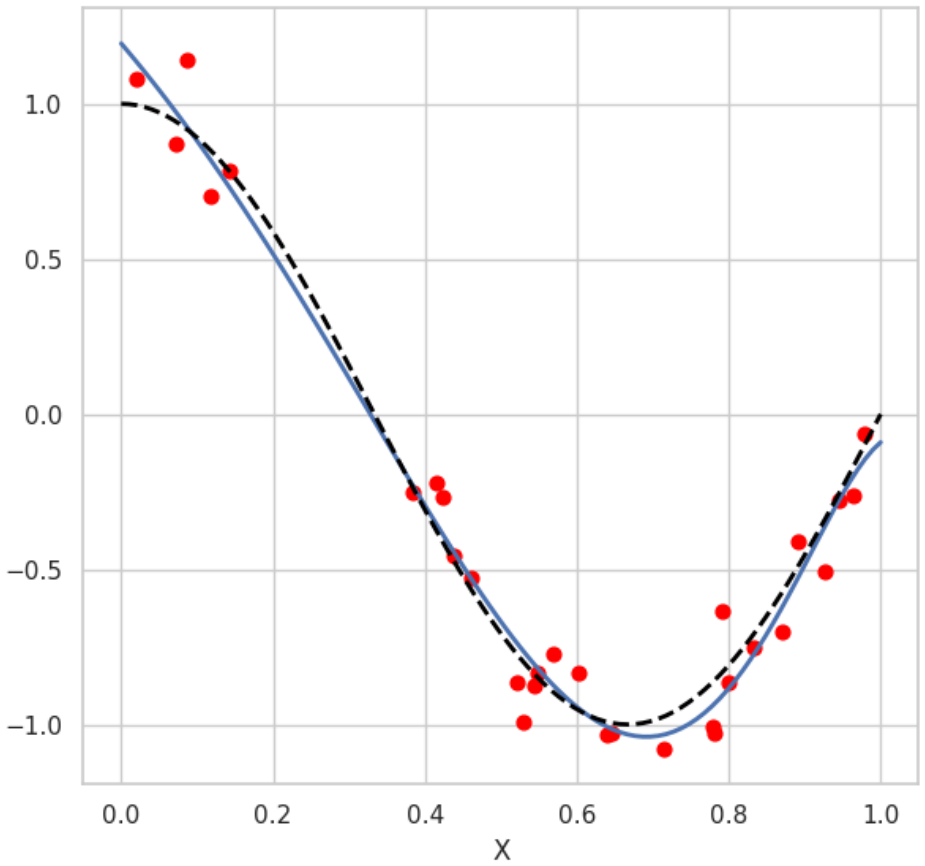


Regularized linear regression (Lasso)



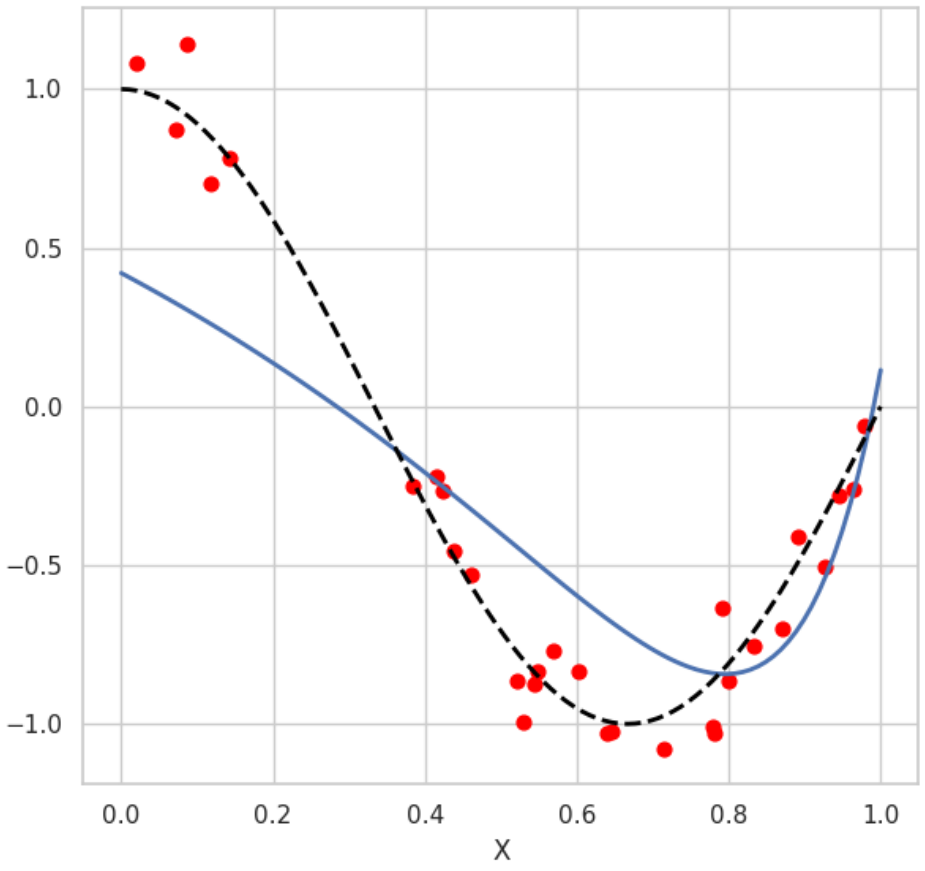


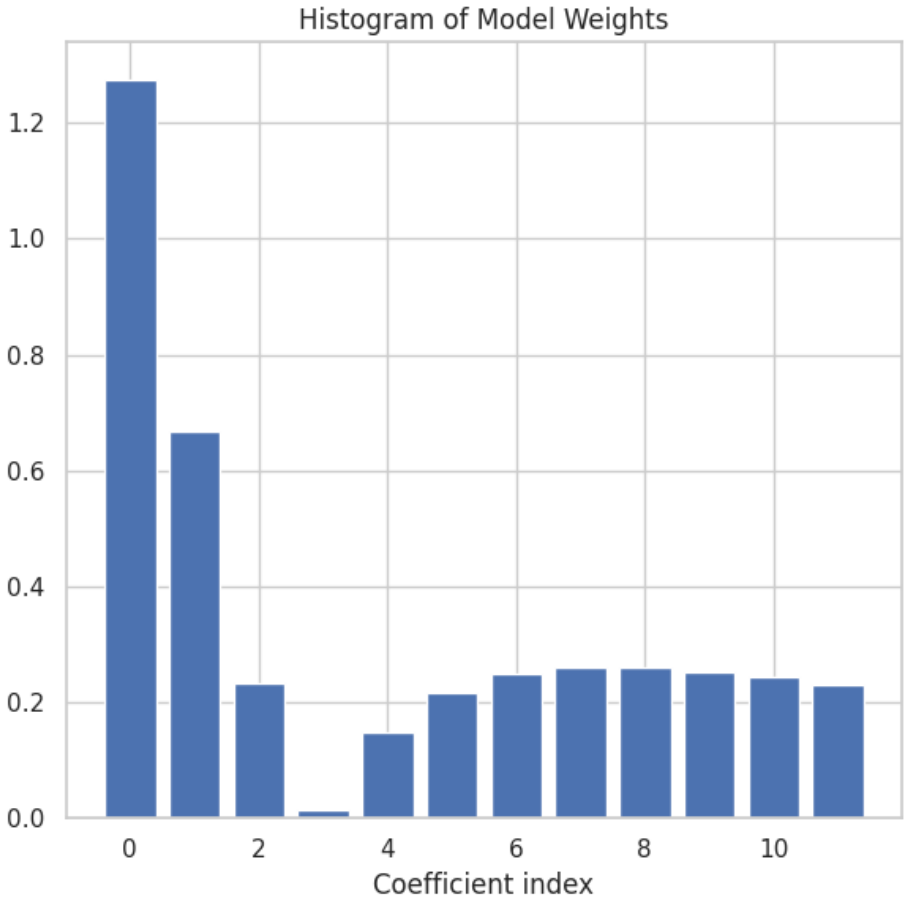
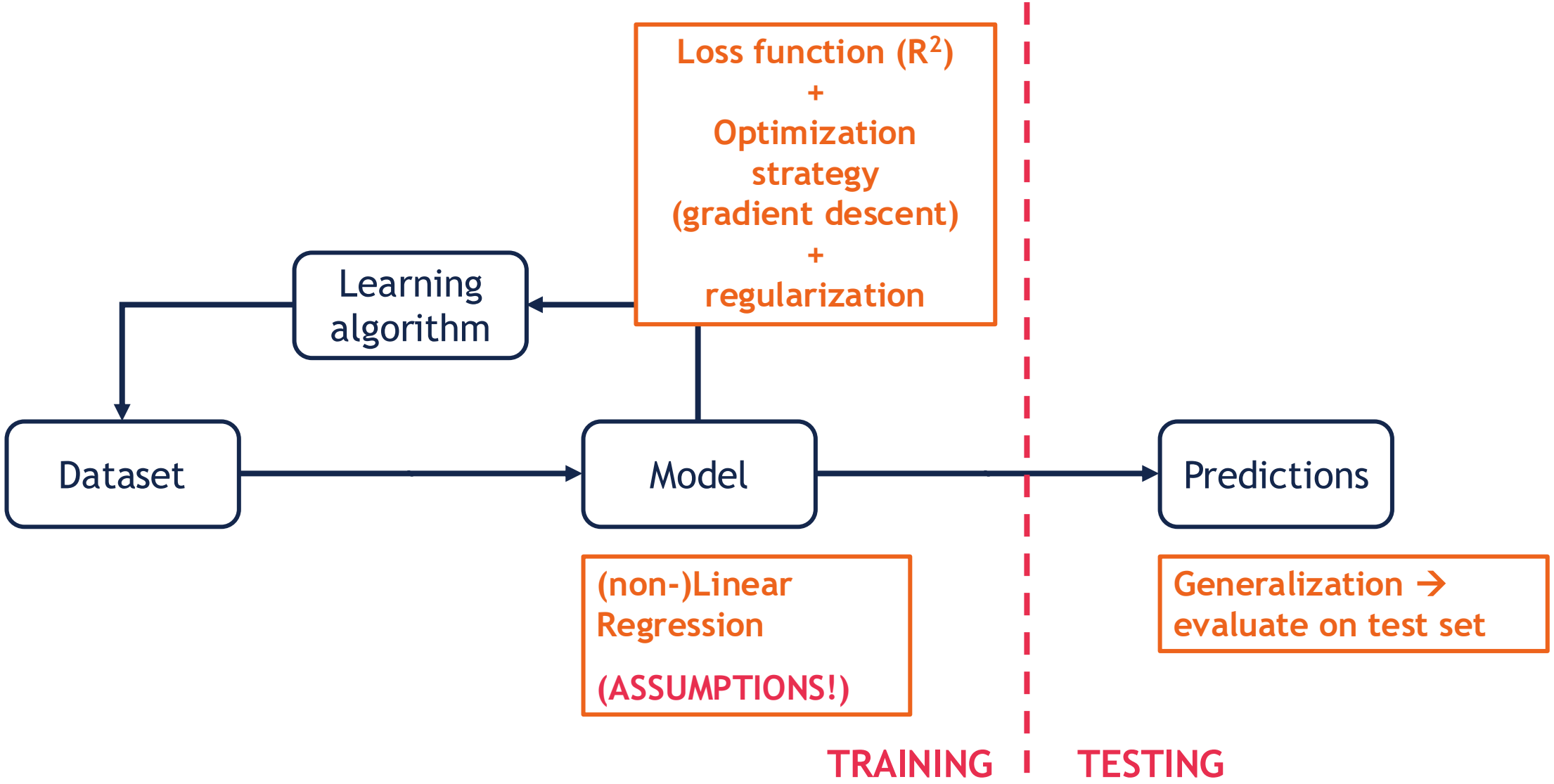
Regularized linear regression (Ridge)



Regularized linear regression (Ridge)

Stronger regularization (λ_2)





Linear regression - recap